

Saekwang Nam

Assistant Professor in the Graduate School of Data Science
 Kyungpook National University, 80 Daehak-ro, Daegu 41566, South Korea
 Web: rottoda.com Email: s.nam@knu.ac.kr

EDUCATION

- 2022 Dr. rer. nat. in Computer Science**, Eberhard Karls University of Tübingen, Germany
 The doctoral research has been conducted at Max Planck Institute for Intelligent Systems, Stuttgart, Germany
 Advisor: Katherine J. Kuchenbecker, Ph.D.
Magna Cum Laude
- 2013 M.S. in Computer Science**, University of California, San Diego, La Jolla, USA
 Academic concentration: Artificial Intelligence
- 2011 B.E. in Human & Mechanical Systems Engineering**, Kanazawa University, Kanazawa, Japan
 Academic concentration: Mechatronics
Summa Cum Laude

EXPERIENCE

Kyungpook National University Mar. 2024 - present
Assistant Professor in the Graduate School of Data Science Daegu, South Korea

I am leading [Robot Touch Data Laboratory](#), where I have focused on

- developing a multimodal tactile sensor for robot hands/grippers.
- building a data-driven touch classification model using a variety of ML models.

Bristol Robotics Laboratory & University of Bristol Nov. 2022 - Feb. 2024
Research Associate in Dexterous Robotics Group Bristol, United Kingdom

I am mainly involved in the [Research Centre for Smart, Collaborative Industrial Robotics](#), funded by EPSRC, where I am working on

- a proprioceptive Fin-ray gripper based on the TacTip principle to estimate deformation, contact point, and depth.
- a ROS-based autonomous robot system featuring a UR-10 robot arm with a Robotiq gripper for object detection, pickup, and placement ([Website](#)).

Max Planck Institute for Intelligent Systems Sep. 2017 - Oct. 2022
Doctoral student Stuttgart, Germany

Title of the dissertation: Understanding the Influence of Moisture on Fingerpad-Surface Interactions

Theme I: Hardware developments for catching physical variables of human finger, for which I

- built an apparatus to measure three-dimensional finger contact forces, contact images, and fingerpad moisture ([link](#), [paper](#)).
- improved the apparatus for better data synchronization, better resolution of fingerprint images, and better ambient temperature controls.

- developed a thin and transparent capacitive-type moisture sensor providing continuous moisture measurements based on laser lithography.

Theme II: Software skills such as image processing, simulations, and optimization

- I implemented a processing algorithm to extract the contact area from contact fingerprint images based on intensity gradient and adaptive threshold methods.
- I constructed a variable optimization system using particle swarm optimization to find the material properties of the human finger through repetitive finger-pressing simulations in COM-SOL Multiphysics ([paper](#)).
- I have been implementing a fast parameter optimization algorithm to quickly extract a human finger's moisture level from optimized electrochemical impedance spectroscopy.

Electronics and Telecommunications Research Institute (ETRI) Apr. 2013 - Aug. 2017
Researcher Daejeon, South Korea

Theme I: Developing transparent and soft actuators, particularly focusing on

- Developing soft actuators based on electro-active polymers
- Figuring out the mechanical properties of dielectric elastomers
- Developing a bio-inspired varifocal lens and a thin-film mirror deformed by electrostatic force ([paper](#)).

Theme II: Connecting users to haptic applications that includes

- Developing a glass-plate actuator vibrating with the electrostatic force created between the stacked glass plates
- Building a graphic user interface to continuously present contact force measured from an optical waveguide-based transparent & flexible sensor array film ([paper](#))
- Generating diverse input signals to haptic devices for the evaluation of a newly developed film-type sensor ([paper](#)).

Samsung Electronics Co., Ltd. Jul. - Sep. 2012
Software intern Suwon, South Korea

I was selected one of the worldwide interns as Samsung's global internship program, targeting excellent foreigners (including Koreans who study overseas) to work at the company in South Korea. In the Device Solutions department, I

- analyzed the traffic congestion of wafer-transferring vehicles in the semiconductor factories based on their log data files,
- computed the vehicle's maximum and average velocity in real-time and updated the information to the database.

DENSOTECHNO Co., Ltd. Aug. 2009
Intern Obu, Japan

- Learned a computer-aided design program (NX),
- Designed a radiator loaded on a Toyota's,
- Conducted simulations under diverse external disturbance conditions,
- Analyzed the areas of the radiator in which the disturbance could cause critical stresses.

AWARDS AND HONORS

2024	Best Workshop Poster Award (with co-authors), ICRA 2024 ViTac Workshop
2021	Honorable Mention for the Best ToH Short Paper Award (with co-authors), World Haptics Conference
2021	Finalist for the Best Video Presentation Award (with co-authors), World Haptics Conference
2020	KIST Europe Scholarship Award (€1,000), Korean Scientists and Engineers Association in Germany
2020	Best Work-in-progress Poster Award (with co-authors), EuroHaptics Conference
2017 - 2022	A full funding for Ph.D. studies, International Max Planck Research School for Intelligent Systems
2015	Creativity and Innovation Award (with co-authors), ETRI
2015	Outstanding Paper Award (with co-authors), ETRI
2015	Outstanding Idea Award, ETRI
2014	Best Demonstration Award (with co-authors), Asia Haptics Conference
2014	The Most Popular Demonstration Award (with co-authors), Asia Haptics Conference
2014	Outstanding Paper Award (with co-authors), Institute of Control, Robotics and Systems domestic conference
2011	Hatakeyama Award, The Japan Society of Mechanical Engineers
2010	Kanazawa-kogyokai Award, Alumni Association in Engineering Departments of Kanazawa University
2009	Special Award, The Japan Society for Precision Engineering Hokuriku-Shinetsu Branch
2006 - 2011	A full funding for undergraduate studies, Korea-Japan Government Scholarship Program for the Students in Science and Engineering Departments (A stipend of ¥125,000/month and tuition fees)

SKILLS

Software	Robot Operating System, MoveIt!, Gazebo, Matlab, Python, LabVIEW, SolidWorks, COMSOL Multiphysics, $\text{\LaTeX}$, C/C++/C#, Unity, Origin, NX
Hardware	UR10, Robotiq gripper, Laser lithography, Metal sputter, Optical microscope, Scanning electron microscope, Laser scanning vibrometer, Phantom haptic devices, Data acquisition board, 3D optical profilometer, Signal generators, 3D printers, Laser cutters
Language	English (<i>fluent</i>), Korean (<i>native</i>), Japanese (<i>excellent</i>), German (<i>beginner(A2)</i>)

RECENTLY COMPLETED COURSES

2021	Machine Learning for and with Dynamical Systems from University of Tübingen Deep Learning from Coursera
2020	Data Literacy from University of Tübingen Probabilistic Machine Learning from University of Tübingen Machine Learning from Coursera
2018	Mathematics for Machine Learning: Multivariate Calculus from Coursera
2018	Mathematics for Machine Learning: Linear Algebra from Coursera

PUBLICATIONS

Unpublished Papers (In Preparation and Under Review)

- [U1] Masoud S. Bahraini, Ella-Mae Hubbard, Tiziana C. Callari, Isaiah Nassiuma, Paul Anandan, Pedro Ferreira, Cong Niu, Xiu Yan, Anas Shrinah, Kerstin Eder, Loong Yi Lee, **Saekwang Nam**, Jonathan Rossiter, Nathan Lepora, Seemal Asif, Iveta Eimontaite, Sarah Fletcher, Phil Webb, Angela Daly, Riccardo Vecellio Segate, Dariusz Ceglarek, Peter Kinnell, Niels Lohse: Human Behaviour and Intention Prediction for Industrial Human-Robot Collaboration: A Survey. *Submitted*.
- [U2] **Saekwang Nam** and Katherine J. Kuchenbecker. The Importance of Sweat in Object Grasping: Why Is Dry Finger Slippery? *In preparation for submission*.
- [U3] **Saekwang Nam**, Katherine J. Kuchenbecker, and the other collaborators. In vivo Transparent Sensor for the Continuous Hydration Measurement of Finger. *In preparation for submission*.

Journal Articles (Including Impact Factors in the year of publications)

- [J1] Taejun Kwon, Jieun Jang, and **Saekwang Nam**. Channel-attention 1d-cnns for real-time collision detection in human-robot interaction via audio spectral features. *International Journal of Control, Automation and Systems*, 23(11):3370–3382, 2025 (*Impact Factor = 2.9*).
- [J2] Loong Yi Lee, Silvia Terrile, **Saekwang Nam**, Tianhao Liang, Nathan Lepora, and Jonathan Rossiter. Fin-a-rays: Expanding soft gripper compliance via discrete arrays of flexible structures. *Soft Robotics*, 2025 (*Impact Factor = 6.1*).
- [J3] Chenghua Lu, Kailuan Tang, Xueming Hui, Haoran Li, **Saekwang Nam**, and Nathan F. Lepora. Stereotactip: Vision-based tactile sensing with biomimetic skin-marker arrangements. *IEEE/ASME Transactions on Mechatronics*, pages 1–13, 2025 (*Impact Factor = 7.3*).
- [J4] Paolo Susini, Giulia Pagnanelli, **Saekwang Nam**, Nathan F. Lepora, and Matteo Bianchi. Data-driven compliance discrimination via biomimetic soft optical tactile sensors: Implementation and benchmarking with a model-based approach. *IEEE Robotics and Automation Letters*, 10(8):8083–8090, 2025 (*Impact Factor = 5.3*).
- [J5] Taejun Kwon and **Saekwang Nam**. A preliminary study on softness optimization using machine learning for the digital twin of soft robots. *Journal of Institute of Control, Robotics and Systems*, 31(5):490–495, 2025 (*SCOPUS indexed*).
- [J6] Haoran Li, **Saekwang Nam**, Zhenyu Lu, Chenguang Yang, Efi Psomopoulou, and Nathan F. Lepora. Biotactip: A soft biomimetic optical tactile sensor for efficient 3d contact localization and 3d force estimation. *IEEE Robotics and Automation Letters*, 9(6):5314–5321, 2024 (*Impact Factor = 5.3*).
- [J7] Seung Koo Park, Sungryul Yun, Geonwoo Hwang, Meejeong Choi, Dong Wook Kim, Jong-Moo Lee, Bong Je Park, **Saekwang Nam**, Heeju Mun, Seongcheol Mun, Jeong Mook Lim, Eun Jin Shin, Ki-Uk Kyung, and Suntak Park. Highly contrastive, real-time modulation of light intensity by reversible stress-whitening of spontaneously formed nanocomposites: application to wearable strain sensors. *Journal of Materials Chemistry C*, 9(27):8496–8505, 2021 (*Impact Factor = 8.067*).

- [J8] **Saekwang Nam** and Katherine J. Kuchenbecker. Optimizing a viscoelastic finite element model to represent the dry, natural, and moist human finger pressing on glass. *IEEE Transactions on Haptics*, 14(2):303–309, 2021. Finalist for the Best Video Presentation Award, Second Honorable Mention for the Best ToH Short Paper Award at World Haptics Conference 2021 (*Impact Factor* = 3.105).
- [J9] **Saekwang Nam**, Yasemin Vardar, David Gueorguiev, and Katherine J. Kuchenbecker. Physical variables underlying tactile stickiness during fingerpad detachment. *Frontiers in Neuroscience*, 14:235, 2020 (*Impact Factor* = 4.677).
- [J10] **Saekwang Nam**, Sungryul Yun, Jae Woong Yoon, Suntak Park, Seung Koo Park, Seongcheol Mun, Bongje Park, and Ki-Uk Kyung. A robust soft lens for tunable camera application using dielectric elastomer actuators. *Soft Robotics*, 5(6):777–782, 2018. PMID: 30156468 (*Impact Factor* = 6.403).
- [J11] Seongcheol Mun, Sungryul Yun, **Saekwang Nam**, Seung Koo Park, Suntak Park, Bong Je Park, Jeong Mook Lim, and Ki-Uk Kyung. Electro-active polymer based soft tactile interface for wearable devices. *IEEE Transactions on Haptics*, 11(1):15–21, 2018 (*Impact Factor* = 2.757).
- [J12] Suntak Park, Bongje Park, **Saekwang Nam**, Sungryul Yun, Seung Koo Park, Seongcheol Mun, Jeong Mook Lim, Yeonghwa Ryu, Seok Ho Song, and Ki-Uk Kyung. Electrically tunable binary phase fresnel lens based on a dielectric elastomer actuator. *Optics Express*, 25(20):23801–23808, 2017 (*Impact Factor* = 3.356).
- [J13] Seung Koo Park, Young-Je Kwark, **Saekwang Nam**, Jaehyun Moon, Dong Wook Kim, Suntak Park, Bongje Park, Sungryul Yun, Jeong-Ik Lee, Byounggon Yu, et al. A variation in wrinkle structures of uv-cured films with chemical structures of prepolymers. *Materials Letters*, 199:105–109, 2017 (*Impact Factor* = 2.687).
- [J14] Sungryul Yun, Suntak Park, **Saekwang Nam**, Bongje Park, Seung Koo Park, Seongcheol Mun, Jeong Mook Lim, and Ki-Uk Kyung. An electro-active polymer based lens module for dynamically varying focal system. *Applied Physics Letters*, 109(14):141908, 2016 (*Impact Factor* = 3.411).
- [J15] Seung Koo Park, Young-Je Kwark, **Saekwang Nam**, Suntak Park, Bongje Park, Sungryul Yun, Jaehyun Moon, Jeong-Ik Lee, Byounggon Yu, and Ki-Uk Kyung. Wrinkle structures formed by formulating uv-crosslinkable liquid prepolymers. *Polymer*, 99:447–452, 2016 (*Impact Factor* = 3.684).
- [J16] **Saekwang Nam**, Suntak Park, Sungryul Yun, Bongje Park, Seung Koo Park, and Ki-Uk Kyung. Structure modulated electrostatic deformable mirror for focus and geometry control. *Optics Express*, 24(1):55–66, 2016 (*Impact Factor* = 3.307).
- [J17] Bong Je Park, Suntak Park, Sungryul Yun, **Saekwang Nam**, and Ki-Uk Kyung. A transparent visuo-haptic input device with optical waveguide based thin film display, sensor and surface actuator. *Sensors and Actuators A: Physical*, 233:47–53, 2015 (*Impact Factor* = 2.201).
- [J18] Sungryul Yun, Suntak Park, Bongje Park, **Saekwang Nam**, Seung Koo Park, and Ki-Uk Kyung. A thin film active-lens with translational control for dynamically programmable optical zoom. *Applied Physics Letters*, 107(8):081907, 2015 (*Impact Factor* = 3.142).
- [J19] Suntak Park, Bong Je Park, Sungryul Yun, **Saekwang Nam**, Seung Koo Park, and Ki-Uk

Kyung. Thin film display based on polymer waveguides. *Optics express*, 22(19):23433–23438, 2014 (*Impact Factor* = 3.488).

- [J20] Sungryul Yun, Suntak Park, Bongje Park, Youngsung Kim, Seung Koo Park, **Saekwang Nam**, and Ki-Uk Kyung. Polymer-waveguide-based flexible tactile sensor array for dynamic response. *Advanced Materials*, 26(26):4474–4480, 2014 (*Impact Factor* = 17.493).
- [J21] Ki-Uk Kyung, Sungryul Yun, Suntak Park, Bongje Park, Seung Koo Park, **Saekwang Nam**, Harsha Prahlad, and Philip von Guggenberg. Flexible visuo-haptic display. *Journal of Korea Robotics Society*, 8:156–163, 2013 (*No Impact Factor Available*).

Peer-Reviewed Conference Papers, posters, and demo presentations

- [C1] Loong Yi Lee, Silvia Terrile, Ajmal Roshan, Tianqi Yue, **Saekwang Nam**, and Jonathan Rossiter. Down the rabbit hole: Exploiting airflow interactions for morphologically intelligent retracting vacuum grippers. In *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, pages 870–875, 2024
- [C2] **Saekwang Nam***, Toby Jack*, Loong Yi Lee, and Nathan F. Lepora. Softness prediction with a soft biomimetic optical tactile sensor. In *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, pages 121–126, 2024
- [C3] **Saekwang Nam**, Loong Yi Lee, Max Yang, Naoki Shitanda, Jonathan Rossiter, and Nathan Lepora. Vision and tactile pose identification for picking a target without collision. Poster presented at “ViTac” workshop held at ICRA 2023 Conference, May 2023
- [C4] **Saekwang Nam**, David Gueorguiev, and Katherine J. Kuchenbecker. Finger contact during pressing and sliding on a glass plate. Poster presented at “Skin mechanics and its role in manipulation and perception” workshop held at EuroHaptics, May 2022
- [C5] **Saekwang Nam** and Katherine J. Kuchenbecker. Sweat softens the outermost layer of the human finger pad: evidence from simulations and experiments. Work-in-progress poster presented at EuroHaptics Conference, September 2020. **Award for the Best Work-in-progress poster**
- [C6] **Saekwang Nam** and Katherine J. Kuchenbecker. Understanding the pull-off force of the human fingerpad. Work-in-progress paper (2 pages) presented at the IEEE World Haptics Conference (WHC), July 2019
- [C7] Maria-Paola Forte, Rachael L’Orsa, Mayumi Mohan, **Saekwang Nam**, and Katherine J. Kuchenbecker. The haptician and the alphamonsters. Student Innovation Challenge on Implementing Haptics in Virtual Reality Environment presented at the IEEE World Haptics Conference, July 2019. Maria Paola Forte, Rachael L’Orsa, Mayumi Mohan, and Saekwang Nam contributed equally to this publication
- [C8] Seongcheol Mun, Sungryul Yun, **Saekwang Nam**, Seung-Koo Park, and Ki-Uk Kyung. Initial progress toward a surface morphable tactile interface. In Shoichi Hasegawa, Masashi Konyo, Ki-Uk Kyung, Takuya Nojima, and Hiroyuki Kajimoto, editors, *Haptic Interaction*, pages 113–114, Singapore, 2018. Springer Singapore
- [C9] **Saekwang Nam**, Suntak Park, Sungryul Yun, Bong Je Park, Seung Koo Park, and Ki-Uk Kyung. Influence on deformed shape of electrostatic deformable mirror by the geometry of electrode. In *Conference on Lasers and Electro-Optics*, page JTh2A.7. Optical Society of America, 2016

- [C10] Suntak Park, Jin Tae Kim, **Saekwang Nam**, Bong Je Park, Sungryul Yun, Seung Koo Park, and Ki-Uk Kyung. Pressure sensor using vertical coupling of optical waveguides. In *Conference on Lasers and Electro-Optics*, page JTu5A.144. Optical Society of America, 2016
- [C11] **Saekwang Nam**, Suntak Park, Sungryul Yun, Bong Je Park, Seung Koo Park, Mijeong Choi, and Ki-Uk Kyung. *Highly Flexible and Transparent Skin-like Tactile Sensor*, pages 187–189. Springer Japan, Tokyo, 2015
- [C12] **Saekwang Nam** and Tetsuyou Watanabe. The development of robot hand capable of expressing the movement of the palm(biorobotics). *Proceedings of the robotics and mechatronics conference in Japan*, 2011:1P1–P04(1–4), May 2011

PATENTS

- [P1] Seung Koo Park, Sung Ryul Yun, Ki Uk Kyung, Mi Jeong Choi, Bong Je Park, Suntak Park, **Saekwang Nam**, Seongcheol Mun, Eun Jin Shin, Jeong Mook Lim, et al. Light transmittance control film and composition for the light transmittance control film, March 30 2023. US Patent 2023/0,103,068
- [P2] Ki-uk Kyung, Bong Je Park, Sang-youn Kim, **Saekwang Nam**, Sun Tak Park, and Sung Ryul Yun. Auto focusing device, May 12 2020. US Patent 10,649,116
- [P3] Ki-uk Kyung, **Saekwang Nam**, Bong Je Park, Sun Tak Park, and Sung Ryul Yun. Imaging apparatus with adjustable lens and method for operating the same, December 3 2019. US Patent 10,495,843
- [P4] Ki Uk Kyung, **Saekwang Nam**, Sung Ryul Yun, Bong Je Park, Sun Tak Park, and Seung Koo Park. Varifocal lens module, July 2 2019. US Patent 10,338,280
- [P5] Ki-uk Kyung, Bong Je Park, Sang-youn Kim, **Saekwang Nam**, Sun Tak Park, and Sung Ryul Yun. Auto focusing device, June 26 2018. US Patent 10,007,034
- [P6] Sun Tak Park, Ki Uk Kyung, Sae Kwang Nam, Bong Je Park, Seung Koo Park, and Sung Ryul Yun. Optical imaging device, June 26 2018. US Patent 10,009,546
- [P7] Sun Tak Park, Ki-uk Kyung, **Saekwang Nam**, Bong Je Park, Seung Koo Park, and Sung Ryul Yun. Image processing apparatus and control method thereof, June 5 2018. US Patent 9,992,433
- [P8] Sung-Ryul Yun, Ki-uk Kyung, Sun-tak Park, Bong-Je Park, and **Saekwang Nam**. Microlens array and method for fabricating thereof, February 6 2018. US Patent 9,885,874
- [P9] Sun Tak Park, Ki Uk Kyung, Sung Ryul Yun, **Saekwang Nam**, and Bong Je Park. Display apparatus and manufacturing method thereof, December 12 2017. US Patent 9,841,621
- [P10] Ki-uk Kyung, Sung-Ryul Yun, Sun-tak Park, Bong-Je Park, and **Saekwang Nam**. Shape-variable optical element, August 29 2017. US Patent 9,746,587
- [P11] Sung Ryul Yun, Ki Uk Kyung, **Saekwang Nam**, Bong Je Park, and Sun Tak Park. Artificial muscle, May 30 2017. US Patent 9,662,197
- [P12] Sun-tak Park, Bong-Je Park, Ki-uk Kyung, Sung-Ryul Yun, and **Saekwang Nam**. Active diffuser for reducing speckle and laser display device having active diffuser, May 2 2017. US Patent 9,638,928

- [P13] Sung Ryul Yun, Ki Uk Kyung, Sun Tak Park, **Saekwang Nam**, and Bong Je Park. Thin active optical zoom lens and apparatus using the same, April 4 2017. US Patent 9,612,362
- [P14] **Saekwang Nam**, Sun Tak Park, Ki Uk Kyung, Bong Je Park, and Sung Ryul Yun. Active reflective lens and apparatus using the same, September 15 2016. US Patent App. 15/065,856
- [P15] Sun Tak Park, Ki Uk Kyung, Bong Je Park, Sung Ryul Yun, and **Saekwang Nam**. Variable fresnel lens, April 12 2016. US Patent 9,310,532
- [P16] **Saekwang Nam**, Ki-uk Kyung, Sung-Ryul Yun, Sun-tak Park, and Bong-Je Park. Shape-variable optical element and optical read/write device including the same, July 28 2015. US Patent 9,091,810
- [P17] Sun Tak Park, Sung Ryul Yun, Bong Je Park, **Saekwang Nam**, and Ki Uk Kyung. Reflective varifocal lens and imaging system including the same, August 20 2015. US Patent App. 14/618,282
- [P18] **Saekwang Nam**, Ki Uk Kyung, Sung Ryul Yun, Bong Je Park, and Sun Tak Park. Tunable lens system, July 23 2015. US Patent App. 14/600,369
- [P19] Ki-uk Kyung, Sung-Ryul Yun, Sun-tak Park, Bong-Je Park, and **Saekwang Nam**. Variable-shape optical element, February 12 2015. US Patent App. 14/316,083

REVIEWER

2025	IEEE Transactions on Mechatronics
2023	IEEE RoboSoft Conference, IEEE Sensors Journal, IEEE Transactions on Haptics (ToH), Intelligent Service Robotics
2021	IEEE World Haptics Conference
2020	IEEE International Conference on Robotics and Automation (ICRA)

TEACHING AND MENTORING

- Jan. 2012 - He had taught **Japanese** to undergraduate students at UCSD for **four quarters**.
- Mar. 2013 - Although he was not a native Japanese speaker, the lecturers kindly gave me continuous chances to have the TA role based on students' evaluations and the Japanese Language Proficiency Test certificate with the highest level.
- November 2020 - He taught the concept of impedance measurements, how to measure the impedance, and how to find the equivalent electrical circuit based on the electrochemical impedance spectroscopy to **Carolyn Anna Both**, a student at Friedrich-Abel-Gymnasium, for one week.
- May 2022 - He introduced the sensing principle of the transparent moisture sensor and ways of impedance measurement and microscope usage to **Tsujii Manon**, a Gymnasium student, for a week.
- Dec. 2022 - **Naoki Shitanda** came to the University of Bristol as a visiting undergraduate student from Kyoto University. During his time there, Saekwang advised on sensor fabrication, the use of a robot arm and the Robot Operating System. Together, they participated in the 2023 RoboSoft manipulation competition.
- May 2023 - He assisted **Toby Jack**, an MSc student in the Biorobotics program at the University of Bristol, in constructing a robotic system designed to measure and evaluate the softness of fruit using a soft tactile sensor.
- Aug. 2023 -
- Nov. 2023 - He is officially supervising **Chenghua Lu**'s doctoral research at the University of present Bristol.

TALKS AND MEDIA HIGHLIGHTS

Jan. 4th 2024	Invited talk in Korea University of Technology and Education
Jan. 3rd 2024	Invited talk in Pusan National University
Dec. 21st 2023	Invited talk in Sookmyung Women's University
Dec. 19th 2023	Invited talk in Electronics and Telecommunications Research Institute
Dec. 18th 2023	Invited talk in Gwangju Institute of Science and Technology
Oct. 18th 2022	Invited talk titled "Current in Robot Fingers" to the Mechanical Engineering Department in Republic of Korea Naval Academy
Aug. 3rd 2022	Invited talk titled "Optimizing a Viscoelastic Finite Element Model to Represent the Dry, Natural, and Moist Human Finger Pressing on Glass" to Prof. Chun's lab. in Korea University Sejong Campus
Dec. 15th 2021	A report titled "The artificial intelligence in the sense of touch: The present and future of haptic technologies" has been published in KIC Europe Issue 2021/12: Artificial intelligence & Robotics (Korean).
May 27th 2021	A method to optimize the parameters in a model using measured data and a case example in modeling finger-pressing behaviors, organized by Korean Scientist and Engineers Association in Germany (Online presentation)
May 24th 2021	Designing, Sensing, and Data-processing for Understanding Human Fingers, hosted by the Embodied Dexterity Group at University of California Berkeley
November 2020	He was invited for the interview with Tech GangJeong, a youtube channel focusing on new technologies. He introduced the procedures of doctoral studies in Max Planck Institute , my research , and current challenges in object grasping by robots .
Oct. 17th 2020	Reasons for the secure object-grasping by fingers, organized by Korean Scientist and Engineers Association in Germany (Online presentation)
October 2020	Max Planck Institute for Intelligent Systems published an article on how well several members in our department participated in the EuroHaptics conference in the COVID-19 outbreak. News of receiving the Best Poster Prize featured in the article.
May 2017	HelloDD published an article about the importance of the Technical Research Personnel Program, an alternative means of conducting mandatory military service in Korea.
January 2017	Chungcheong Today published Saekwang's opinion about the importance of developing path-breaking technologies.
June 2014	Many broadcasting companies in South Korea released news about the development of transparent tactile sensor array (KBS , Daejeon MBC , TJB , Yonhap News , YTN , and Arirang).

REFERENCES

Name		Position and affiliation	E-mail
Nathan F. Lepora		Professor at University of Bristol and Bristol Robotics Laboratory	n.lepora@bristol.ac.uk
Katherine Kuchenbecker	J.	Director of Haptic Intelligence Department at Max Planck Institute for Intelligent Systems	kjk@is.mpg.de
Ki-Uk Kyung		Professor in Mechanical Engineering at KAIST	kyungku@kaist.ac.kr